

NONABELIAN COMPOSITION FACTORS OF FINITE GROUPS WHOSE MAXIMAL SUBGROUPS ARE COMPLEMENTED

RICHIE SATER

ABSTRACT. Assume the classification of finite simple groups and the external factorization consequences recorded in Appendix C. Let G be a finite group in which every maximal subgroup M has an exact complement: $G = MK$ and $M \cap K = 1$ for some $K \leq G$. We prove that every nonabelian composition factor of G is isomorphic to

$$\mathrm{PSL}_2(7), \quad \mathrm{PSL}_2(11), \quad \text{or} \quad \mathrm{PSL}_5(2),$$

and that all three possibilities occur. Subject to those inputs, this answers Kourovka Problem 18.68. Two load-bearing LPS source matches remain at the explicit verification boundary described in the introduction and Appendix C.

The central difficulty is that the complement property does not pass to sections. We replace that invalid inheritance step by a section-safe construction: a chosen composition factor S yields a monolithic quotient with socle S^k , and its almost-simple coordinate action lifts to a faithful primitive action on a Cartesian power. A regular complement would then force a core-free transitive subgroup in the coordinate action. Factor-free, prime-elusive, and socle-valuation obstructions reduce this coordinate obligation to published exhaustive factorization classifications. The only infinite screen survivor, $\mathrm{PSp}_4(2^f)$, is eliminated by a prime-degree subfield action and a 2-adic argument. Complete tables of marks certify only fifteen named small almost-simple cases; they are not used for any infinite-family conclusion.

1. INTRODUCTION

Say that a finite group G has property **CMP** if every maximal subgroup $M < G$ has a subgroup $K \leq G$ such that

$$(1) \quad G = MK, \quad M \cap K = 1.$$

The complement is not assumed normal. Kourovka Problem 18.68 asks for the nonabelian composition factors of a nonsoluble group with this property [9, Problem 18.68].

Levchuk and Likharev proved that the nonabelian simple groups having **CMP** are precisely $\mathrm{PSL}_2(7)$, $\mathrm{PSL}_2(11)$, $\mathrm{PSL}_5(2)$ [13, Theorem 1 and Lemmas

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4, 6]. Maslova obtained the same composition-factor list under the stronger hypothesis that every maximal subgroup is a Hall subgroup [19, Theorem 1]. Maslova and Revin subsequently described groups under that hypothesis and proved that it implies CMP [20, Theorems 1–2]; the implication is not reversible, so those results do not answer the present question. The complete classification of exact factorizations of almost-simple groups [14] likewise addresses the one-coordinate setting, not an ambient monolithic group with socle S^k . Recent work of Monakhov–Sokhor concerns monotone index sequences in maximal chains [21], rather than exact complements or the nonabelian composition factors considered here. A literature search updated on 31 August 2026 found no published solution of Problem 18.68; this negative search is not a claim of absolute priority. The July 2026 edition of the Kourovka Notebook still prints the problem without a solution annotation.

The structural setting has established roots. Aschbacher–Scott organize maximal subgroups through minimal normal subgroups [1]; Kovács studies maximal subgroups of composite finite groups [11] and primitive subgroups of wreath products in product action [12]. We do not claim the chief-factor or product-action frameworks themselves. The contribution here is their CMP-specific synthesis: quotient closure places an arbitrary composition factor in a monolithic group, a selected coordinate action lifts to that same quotient, and three complementary obstructions close the resulting classification screen.

The obstacle is inheritance. Property CMP passes to quotients but not to normal subgroups, hence not automatically to sections. For example, D_8 has CMP, while its normal cyclic subgroup C_4 does not. Thus one cannot attach CMP directly to a nonabelian composition factor and quote the simple-group theorem.

Our CMP-specific replacement is the section-safe product-action lift in Theorem 2.3. Starting from a chosen composition factor S , we pass only to a quotient L whose unique minimal normal subgroup is S^k . The group induced by a coordinate is almost simple, $S \leq X \leq \text{Aut}(S)$. A suitable primitive action $X \curvearrowright \Delta$ then manufactures a faithful primitive action $L \curvearrowright \Delta^k$. If L retained CMP, its point stabilizer would have a regular complement. Under the first daggered consequence attributed to Liebeck–Praeger–Saxl (LPS), that regular subgroup forces a coordinate factorization, where three obstruction modes can be applied.

Theorem 1.1. *Assume CFSG and the external classification and factorization statements listed in Appendix C, including the two LPS consequences marked there with a dagger. Let G be a finite group with CMP. Every non-abelian composition factor of G is isomorphic to one of*

$$\text{PSL}_2(7), \quad \text{PSL}_2(11), \quad \text{PSL}_5(2).$$

Each of the three possibilities occurs.

The two daggered inputs are LPS (2000), Corollary 3(iv), for the product-action coordinate consequence, and LPS (1990), Theorem D with Remark 2, for the large alternating branch. Their exact primary texts were not independently reopened during the source traces through 31 August 2026. This paper therefore consumes only the precise consequences stated in [Appendix C](#); if either source has narrower hypotheses, the corresponding branch is unchecked.

The reusable CMP-specific part of the proof is the lift from a quotient-closed maximal-subgroup property and the three-mode coordinate toolkit, not the underlying product-action framework or the subsequent enumeration of families supplied by the classification of finite simple groups (CFSG). We therefore prove the structural reduction in [Section 2](#) and [Section 3](#), give one instructive use of each obstruction in [Section 4](#), and state the exhaustive classification interface in [Section 5](#). The main theorem is assembled in [Section 6](#). Repetitive family checks and finite certificate details are placed in [Appendix A](#) and [Appendix B](#); [Appendix C](#) records the exact external inputs and trust boundary.

2. PROOF ARCHITECTURE AND THE SECTION-SAFE LIFT

The logical route from a composition factor to a contradiction is

$$(2) \quad \boxed{\begin{array}{l} S \rightsquigarrow L \text{ with } \text{Soc}(L) = S^k \rightsquigarrow X = N_L(S_1)/C_L(S_1), \\ L \hookrightarrow X \wr P \curvearrowright \Delta^k \rightsquigarrow L_\omega \text{ with no complement.} \end{array}}$$

Here $\rightsquigarrow$ denotes a construction, not a homomorphism. In particular, X is the group induced by a coordinate stabilizer; the proof never asserts a quotient map $L \rightarrow X$. This distinction is what makes the argument safe when CMP fails to pass to sections.

2.1. The valid inheritance step.

Lemma 2.1 (quotient closure). *If G has CMP and $N \trianglelefteq G$, then G/N has CMP.*

Proof. Let $\overline{M}$ be maximal in G/N , and let M be its full preimage. Choose $K \leq G$ satisfying (1). Then $G/N = \overline{M}(KN/N)$. If $kN \in \overline{M}$ with $k \in K$, then $k \in M$, and hence $k = 1$. Thus KN/N is an exact complement to $\overline{M}$. $\square$

Proposition 2.2 (monolithic reduction). *Let S be a nonabelian composition factor of a finite CMP-group G . There is a quotient L of G such that*

- (1) L has CMP;
- (2) $\text{Soc}(L) = N \cong S^k$ for some $k \geq 1$, and N is the unique minimal normal subgroup of L ;
- (3) $C_L(N) = 1$.

Proof. Choose a chief factor $A/B \cong S^k$ of G , and define

$$C/B = C_{G/B}(A/B), \quad L = G/C, \quad N = AC/C.$$

Since A/B is centerless, $A \cap C = B$, and hence $N \cong A/B$. Chiefness makes N a minimal normal subgroup of L . If gC centralizes AC/C , then $[g, A] \leq A \cap C = B$, so $g \in C$ by the definition of $C/B = C_{G/B}(A/B)$. Thus the conjugation action of L on N is faithful and $C_L(N) = 1$. If R were another minimal normal subgroup, then $[R, N] \leq R \cap N = 1$, forcing $R \leq C_L(N) = 1$. Thus N is unique. Finally, L has CMP by [Lemma 2.1](#). $\square$

The example $C_4 \trianglelefteq D_8$ shows why the preceding proposition, rather than section inheritance, is the correct starting point. The next construction turns its entire direct-power socle into a coordinate problem.

2.2. Manufacturing the product action. Write

$$N = S_1 \times \cdots \times S_k \cong S^k.$$

The action of L on the factors is transitive, since the product of the factors in any orbit is normal in L . Set

$$(3) \quad X = N_L(S_1)/C_L(S_1),$$

identifying S_1 with its inner automorphism group. Then $S \leq X \leq \text{Aut}(S)$. The standard wreath embedding gives

$$(4) \quad L \hookrightarrow X \wr P \leq X \wr \text{Sym}(k),$$

where P is transitive, N is the base socle S^k , and the group induced on a coordinate is X . An element in the kernel fixes every factor and induces the identity automorphism on each S_i , so it centralizes N . Thus the kernel lies in $C_L(N) = 1$, and the embedding is faithful.

Let $H < X$ be core-free and maximal. Put $\Delta = X/H$ and $V = H \cap S$, and assume $V \neq 1$. Since S is normal and is not contained in H , we have $X = SH$; in particular, S is transitive on Δ . Via (4), let L act on Δ^k in product action.

Theorem 2.3 (section-safe product-action lifting). *The action of L on Δ^k is faithful and primitive. If $\omega = (H, \dots, H)$ and $M = L_\omega$, then $L = MS^k$ and $M \cap S^k = V^k$.*

Here we apply standard wreath-product and Scott-lemma ingredients. The proof tests an intermediate overgroup through its intersection with S^k , then uses Scott's lemma to show that the only possible strips have length one.

Proof. Let $\omega = (H, \dots, H)$, $M = L_\omega$, and $N = S^k$. The group N is transitive, so $L = MN$ and $M \cap N = V^k$.

Suppose $M \leq J \leq L$, and put $D = J \cap N$. Then $J = MD$, $D \trianglelefteq M$, and $V^k \leq D$. The image of M on the coordinates is transitive: multiply an element of L by a suitable base element to make it fix ω without changing

its top permutation. Moreover, the subgroup of M stabilizing coordinate i induces H there. For a prescribed $h \in H$, lift h through $N_L(S_i) \rightarrow X$. Its i -coordinate already fixes the coset H , and the transitivity of each other factor permits multiplication by suitable elements of S_j , one for each $j \neq i$, that return all remaining coordinates to H . The corrected lift lies in M and still induces h on S_i .

Let D_i be the projection of D to $S_i \cong S$. It contains V and is normalized by H . Maximality of H implies either $D_i = V$ or $HD_i = X$. In the second case

$$D_i = (H \cap S)D_i = (HD_i) \cap S = X \cap S = S.$$

Coordinate transitivity gives the same alternative for every i . If all projections are V , then $D = V^k$ and $J = M$. Otherwise D is subdirect in S^k . By Scott's lemma [22, Lemma, p. 328] it is a direct product of full diagonal strips. Since $V^k \leq D$ and $V \neq 1$, every strip has length one, so $D = N$ and $J = L$. Thus M is maximal. Faithfulness follows from core-freeness of H and (4). $\square$

The theorem is the decisive removal of the unbounded exponent k . It constructs a new action of the same quotient L ; it does not assume that the original primitive action supplied by CMP already has product-action type. The next section explains how the existence of a complement in this manufactured action constrains the one-coordinate group X .

3. THE COORDINATE-OBSTRUCTION TOOLKIT

Let $H < X$ be as in Theorem 2.3, put $\Delta = X/H$, and write $V = H \cap S \neq 1$. A complement to the lifted point stabilizer is a regular subgroup, so it can be ruled out at any of three increasingly flexible levels.

Throughout, X/H denotes the set $\{Hx : x \in X\}$ of right cosets, on which X acts by right multiplication. Dotted Atlas notation such as $M_{12.2}$ denotes an extension without asserting that it splits; a colon denotes a split extension, and subscripts distinguish Atlas isomorphism types.

Mode	Coordinate statement	Lifting mechanism
factor-free	no core-free $C < X$ satisfies $X = HC$	the daggered LPS Corollary 3(iv) consequence, with a direct almost-simple argument for $k = 1$
prime-elusive	$p \mid \Delta $ and every element of order p fixes a point	fixed points lift to the full wreath product on Δ^k
socle valuation	S is p -elusive and $v_p(\Delta) > v_p(X : S)$	a regular subgroup has a nontrivial p -element in its intersection with S^k

The formal statements below isolate exactly what the classification appendix must supply.

Lemma 3.1 (regular-action reformulation). *A subgroup $K \leq G$ satisfies (1) if and only if it is regular on the right cosets of M . If M is maximal and core-free, this is a faithful primitive action.*

Proof. Transitivity is equivalent to $G = MK$, and the stabilizer in K of the coset M is $M \cap K$. $\square$

Call (X, Δ) *factor-free* if it has no core-free transitive subgroup; equivalently, there is no core-free $C < X$ with $X = HC$.

Proposition 3.2 (coordinate obstruction). *If $(X, X/H)$ is factor-free, then L does not have CMP.*

Proof. For $k = 1$, we have $L = X$. If a regular subgroup R existed, then R would be transitive and $R \cap H = 1$. Its core in X , if nontrivial, would contain the socle S , as X is almost simple. But then $S \leq R$, contrary to $1 \neq H \cap S \leq H$ and $R \cap H = 1$. Thus R would be core-free, contradicting factor-freeness. Let $k \geq 2$. The action in Theorem 2.3 is a primitive product action with almost-simple component $(X, X/H)$: indeed, its socle stabilizer is V^k , which is nontrivial and is not subdirect in S^k . The first daggered source assumption is the consequence attributed to Corollary 3(iv) of [17]: if such a group has a regular subgroup, then its component action has a core-free transitive subgroup. Hence no regular subgroup exists. But if L had CMP, its maximal point stabilizer would have a complement, regular by Lemma 3.1, a contradiction. $\square$

This is the step that removes the unbounded exponent k : it converts the entire product-action problem into one almost-simple factorization screen.

We first state the mechanism for applying maximal-factorization tables.

Lemma 3.3 (factor screen). *Let $1 < V < S$, and assume that the S -conjugacy class of V is invariant under X . Choose a maximal subgroup $H < X$ containing $N_X(V)$. Then H is core-free and $X = HS$. If X/H is not factor-free, there are $S \leq Y \leq X$ and a nontrivial maximal factorization $Y = AB$ such that, after interchanging the factors,*

$$V \leq A \cap S.$$

Proof. Class invariance gives $N_X(V)S = X$, so no proper overgroup of $N_X(V)$ contains S . Hence H is core-free and $HS = X$.

Suppose $C < X$ is core-free and $X = HC$. Put $Y = CS$. Then $Y = (H \cap Y)C$, and both factors supplement S in Y . Consequently all their proper maximal overgroups are core-free. Enlarging them gives a maximal factorization $Y = AB$, with $V \leq H \cap S \leq A \cap S$. $\square$

Definition 3.4. An action of Z on Λ is *p-elusive* if $p \mid |\Lambda|$ and every element of order p in Z fixes a point.

Lemma 3.5 (wreath lifting). *If X is p-elusive on Δ , then $X \wr \text{Sym}(k)$ is p-elusive on Δ^k . In particular, no subgroup of this wreath product is regular on Δ^k .*

Proof. Let $w = (x_1, \dots, x_k)\sigma$ have order p . On a fixed coordinate of σ , the corresponding x_i has order 1 or p , and hence fixes a point. On a p -cycle of σ , the cycle product of the x_i is 1, since $w^p = 1$; the fixed-tuple equations can therefore be solved successively around the cycle. Thus w fixes a tuple.

If a subgroup were regular, Cauchy's theorem would give it a nonidentity element of order p , because its order is $|\Delta|^k$. That element would fix a point, a contradiction. $\square$

Lemma 3.6 (socle-valuation lifting). *Suppose S is p -elusive on $\Delta = X/H$. Let*

$$a = v_p(|\Delta|), \quad o = v_p(|X : S|).$$

If $a > o$, then every subgroup $L \leq X \wr \text{Sym}(k)$ containing S^k contains no regular subgroup on Δ^k .

Proof. Suppose $R \leq L$ is regular and put $Q = R \cap S^k$. Since $R/Q \hookrightarrow L/S^k$,

$$v_p(|R : Q|) \leq ko + v_p(k!).$$

As $v_p(|R|) = ka$,

$$v_p(|Q|) \geq k(a - o) - v_p(k!) > 0,$$

because $a - o \geq 1$ and $v_p(k!) < k$. An element of order p in Q fixes a tuple coordinatewise, contrary to regularity. $\square$

4. REPRESENTATIVE OBSTRUCTION PROOFS

This section gives one complete example of each mode. The point is to show how the toolkit works; the repetitive family coverage is deferred to [Appendix A](#).

4.1. Prime-elusive: two-subsets for A_5 .

Proposition 4.1. *For $A_5 \leq X \leq S_5$, the action of X on the ten 2-subsets of $\{1, \dots, 5\}$ is 2-elusive. Its point stabilizer is core-free and maximal, and has nontrivial intersection with A_5 .*

Proof. Write H for the stabilizer of a 2-subset. It is maximal in S_5 , and $H \cap A_5$ is maximal in A_5 ; the actions are therefore primitive and faithful. Every involution in S_5 , and hence in either coordinate group, contains a 2-cycle and fixes its support as a 2-subset. Since $2 \mid 10$, the action is 2-elusive. The subgroup $H \cap A_5$ contains a double transposition and is nontrivial. $\square$

By [Lemma 3.5](#), this single fixed-point observation excludes a regular complement on every Cartesian power.

4.2. Factor-free: six-subsets for large alternating groups.

Proposition 4.2. *Let $n \geq 13$, let $A_n \leq X \leq S_n$, and let H stabilize a 6-subset. Then H is core-free and maximal, $H \cap A_n \neq 1$, and X/H is factor-free.*

Proof. The stabilizer is maximal because $6 < n/2$. The second daggered source assumption, attributed to the alternating-group part of the LPS maximal-factorization theorem [16, Theorem D and Remark 2], is that an intransitive factor in a nontrivial maximal factorization stabilizes a j -subset with $j \leq 5$, apart from listed cases of degrees 6, 8, 10. Consequently no core-free transitive subgroup can factor X with the 6-set stabilizer. Core-freeness follows because the only nontrivial normal subgroup of either coordinate group contains A_n , whereas the stabilizer does not; its intersection with A_n contains $A_6 \times A_{n-6}$. $\square$

The factor screen converts the published list into a containment test: the chosen subgroup type never appears on either side of an allowed maximal factorization.

4.3. Socle valuation: the infinite symplectic survivor.

Proposition 4.3. *Let $S = \mathrm{PSp}_4(2^f)$, $f \geq 2$, and $S \leq X \leq \mathrm{Aut}(S)$. There is a core-free maximal subgroup $H < X$ with $H \cap S \neq 1$ such that S is 2-elusive on X/H and*

$$v_2(|X : H|) > v_2(|X : S|).$$

Consequently the lifted product action has no regular subgroup for any socle multiplicity $k \geq 1$.

Proof. Let $q = 2^f$. Choose a prime $r \mid f$, put $d = f/r$, $q_0 = 2^d$, and let

$$V = \mathrm{Sp}_4(q_0) < S$$

be the standard subfield subgroup. It is maximal in S , since the field-extension degree r is prime [2, Table 8.14]; see also [3, Proposition 4.2 and Table 3].

We check explicitly that this one choice works for every field-only and graph-outer coordinate group $S \leq X \leq \mathrm{Aut}(S)$. Realize S as the fixed points of the standard Frobenius φ^f in the adjoint algebraic group of type B_2 over $\mathbb{F}_2$. The exceptional graph-field endomorphism ρ satisfies $\rho^2 = \varphi$, as in the setup preceding Lemma 2.1 and Theorem 4(a) of [7], while $\mathrm{Aut}(S) = \langle S, \rho \rangle$ by [7, Lemma 2.1]. Since ρ commutes with φ , it normalizes the fixed-point subgroup $V = \mathrm{Sp}_4(2^d)$. Thus the S -class of V is invariant under all of $\mathrm{Aut}(S)$.

Put $H = N_X(V)$. Class invariance gives $X = SH$, and $H \cap S = N_S(V) = V$. Moreover, H is maximal in X . Indeed, if $H < K < X$, then maximality of V in S gives $K \cap S = V$ or S . In the first case, $V = K \cap S \trianglelefteq K$, so $K \leq N_X(V) = H$; in the second, $S \leq K$ and $X = SH \leq K$. Both are

contradictions. Finally, H is core-free: every nontrivial normal subgroup of the almost-simple group X contains S , whereas $H \cap S = V < S$.

Every involution class of S meets V . Here are the details needed at the prime-field endpoint $q_0 = 2$. By Lemmas 2.1–2.2 and Corollary 2.3 of [18], the three involution classes of S meet

$$O_4^+(q) \cong (\mathrm{SL}_2(q) \times \mathrm{SL}_2(q)) \rtimes \langle \tau \rangle,$$

where τ interchanges the factors, and conjugacy inside this orthogonal group agrees with conjugacy in S . If u represents the unique nonidentity involution class in $\mathrm{SL}_2(q)$, the three orthogonal classes have representatives $(u, 1), (u, u), \tau$. We may take $u \in \mathrm{SL}_2(2)$, so all three representatives lie in $O_4^+(2) \leq \mathrm{Sp}_4(2) \leq V$. Hence S is 2-elusive on X/H .

Since $X = SH$, the degree and its 2-part are

$$(5) \quad |X : H| = |S : V| = \frac{q^4(q^2 - 1)(q^4 - 1)}{q_0^4(q_0^2 - 1)(q_0^4 - 1)}, \quad a = v_2(|X : H|) = 4(f - d).$$

The same outer-automorphism description gives $|X : S| \mid 2f$, so

$$o = v_2(|X : S|) \leq 1 + v_2(f) < 2f \leq 4(f - d) = a.$$

Here $v_2(f) \leq f - 1$, and $f - d = f(1 - 1/r) \geq f/2$. Thus Lemma 3.6 applies uniformly. In particular, when $q = 4$ and $X = \mathrm{Aut}(S)$, the corrected subgroup has $|H| = 2880$, $|H \cap S| = 720$, and degree 1360; these values are also reproduced in Appendix B. $\square$

The valuation proof illustrates why a screen survivor is not a gap: the weaker socle-level fixed-point statement becomes decisive once the outer 2-part is smaller than the degree 2-part. The Appendix A audit shows that no other infinite family survives the first two modes.

5. HOW THE CLASSIFICATION CLOSES

Proposition 5.1. *Let S be a nonabelian finite simple group not isomorphic to $\mathrm{PSL}_2(7)$, $\mathrm{PSL}_2(11)$, or $\mathrm{PSL}_5(2)$, and let $S \leq X \leq \mathrm{Aut}(S)$. There is a core-free maximal subgroup $H < X$, with $H \cap S \neq 1$, for which one of the following holds:*

- (1) X/H is factor-free;
- (2) X is p -elusive on X/H for some prime p ;
- (3) $S = \mathrm{PSp}_4(2^f)$, $f \geq 2$, and the hypotheses of Lemma 3.6 hold with $p = 2$.

Proof. The common proof template is to choose an automorphism-invariant subgroup $1 < V < S$, place $N_X(V)$ in a core-free maximal subgroup H , and apply Lemma 3.3. The following roadmap identifies every family, its exceptional treatment, and the exact appendix location where the containment obligation is checked.

CFSG family	common screen and exceptions	complete audit
alternating	small degrees by explicit elusive actions or five finite certificates; $n \geq 13$ by 6-subsets	Proposition 4.1 and Proposition 4.2 ; Appendix A.1
classical	invariant parabolics, torus normalizers, or geometric subgroups; $L_3(4)$ is finite and even $\mathrm{P}\mathrm{Sp}_4$ uses valuation	Proposition 4.3 ; Appendix A.2 and Appendix B
exceptional Lie type	defining-characteristic Sylow or graph-stable parabolic intersections	Appendix A.3
sporadic	Atlas maximal subgroups checked against Giudici's exhaustive factorization tables; M_{24} is 2-elusive	Appendix A.4

The alternating, classical, exceptional, and sporadic groups exhaust the nonabelian finite simple groups. The cited appendix rows include every outer coordinate group and low-parameter isomorphism. Together with the three representative proofs, they establish the proposition. $\square$

The classification is used only through this proposition. Thus a first reading may skip [Appendix A](#): the structural reason for the result is already visible, while every family obligation remains traceable.

6. PROOF OF THE MAIN THEOREM

Proof of [Theorem 1.1](#). Let S be a nonabelian composition factor of G . Apply [Proposition 2.2](#) to obtain a CMP-quotient L with unique self-centralizing socle S^k , and define its coordinate group X by [\(3\)](#).

Suppose S is not one of the three groups in the conclusion. Choose the maximal subgroup $H < X$ supplied by [Proposition 5.1](#). By [Theorem 2.3](#), the same group L has a faithful primitive action on $(X/H)^k$. If the action X/H is factor-free, [Proposition 3.2](#) contradicts $\mathrm{CMP}(L)$. If it is p -elusive, [Lemma 3.5](#) shows that the primitive action has no regular subgroup, again contradicting $\mathrm{CMP}(L)$. In the final symplectic case, the same contradiction follows from [Lemma 3.6](#). Therefore S is one of the three displayed groups.

Conversely, Levchuk–Likharev prove that each of the three simple groups has CMP [[13](#), Lemmas 4 and 6]; taking the group itself shows that each occurs as a composition factor. $\square$

7. CONCLUSION

The proof teaches how to constrain a composition factor when a maximal-subgroup property is quotient-closed but not section-closed. The monolithic quotient keeps the ambient hypothesis intact, while [Theorem 2.3](#) converts its

direct-power socle into a family of coordinate actions. This use of established monolithic and product-action machinery is CMP-specific: it replaces an invalid inheritance assertion by a primitive action whose point stabilizer is still a maximal subgroup of the original quotient.

The essential hypotheses are visible in the proof. Quotient closure is needed to reach L ; the unique self-centralizing socle gives a faithful wreath embedding; a nontrivial coordinate intersection $H \cap S$ forces primitivity; and the maximal-subgroup property must turn the lifted point stabilizer into a regular subgroup. Other quotient-closed properties that force such a regular or suitably transitive partner may admit a similar reduction, but no general metatheorem is claimed here.

The price of the final list is classification dependence. The product-action lift and the three obstruction lemmas do not themselves identify all simple socles; the audited published lists, together with the two explicit LPS source assumptions, supply that coverage. A classification-free proof, or an axiomatization with a second application, remains a natural direction.

Evidence layer	Exact role in Theorem 1.1
ordinary mathematics	quotient closure, monolithic reduction, the product-action lift, all three obstruction lemmas, containment deductions, the $\mathrm{PSp}_4(2^f)$ subfield and valuation argument, and theorem assembly
published classifications	CFSG family exhaustiveness, the audited maximal-factorization lists, and the stated subfield, automorphism, and involution-class inputs
explicit source assumptions	the two precise consequences attributed to LPS (2000), Corollary 3(iv), and LPS (1990), Theorem D with Remark 2
GAP/TomLib	exact factor-free predicates for five small alternating coordinate groups and ten groups with socle $\mathrm{PSL}_3(4)$, plus nine separately labelled sporadic cross-checks
Python integrity checks	pinned serialization, inventory, hashes, and arithmetic; they do not repeat TomLib's intersection search
GAP/AtlasRep	a non-proof $q = 4$ regression for the symplectic construction; it is not used to prove an infinite-family statement
formal proof assistants	none; the theorem is not claimed to have a kernel-checked formal proof

The layers compose in one direction: the ordinary reduction produces a coordinate obligation; published classifications make the infinite screen exhaustive; complete tables of marks settle only the fifteen named finite residues. [Appendix C](#) gives theorem numbers and hypothesis matches, and [Appendix B](#) gives the finite proposition checked by the software.

APPENDIX A. THE EXHAUSTIVE COORDINATE-FAMILY AUDIT

This appendix proves the family assertions consumed by [Proposition 5.1](#). Each subsection states the common screen, identifies exceptional coordinate groups, and points to the exact published classification locus. The body retains the three arguments that teach the method; this appendix retains the exhaustive case obligations.

A.1. Alternating groups. For $n \neq 6$, the coordinate group is A_n or S_n . The following actions are p -elusive:

Socle	action and degree	prime
A_5	2-subsets, 10	2
$A_6, X = A_6$	natural, 6	2
$A_6, X = S_6$	unordered 3 + 3 partitions, 10	2
A_8	2-subsets, 28	2
A_9	2-subsets, 36	2
A_{10}	4-subsets, 210	2
A_{11}	3-subsets, 165	3
A_{12}	2-subsets, 66	2

These assertions follow from the cycle supports. An involution fixes the support of one 2-cycle. In degree 10, every involution also has an invariant 4-set: use two 2-cycles, or one 2-cycle and two fixed points. An element of order 3 in degree 11 fixes the support of one 3-cycle. For the S_6 partition action, the cycle types $2, 2^2, 2^3$ respectively preserve a 3-set or interchange it with its complement.

The three outer extensions of A_6 not equal to S_6 , and the two groups A_7, S_7 , have factor-free maximal subgroups certified in [Appendix B](#). Their orders and indices are

X	$A_{6.2_2}$	$A_{6.2_3}$	$A_{6.2^2}$	A_7	S_7
$ H $	16	20	32	72	144
$ X : H $	45	36	45	35	35.

For $n \geq 13$, the factor-free action on 6-subsets is proved in [Proposition 4.2](#).

A.2. Classical groups. We use Theorem 2.15 and Appendix Tables A.1–A.7 of Xia–Li [15], a restatement of the LPS classification of nontrivial maximal factorizations with classical socle, together with the two corrected $P\Omega_8^+(4)$ and $P\Omega_8^+(16)$ rows of Gill–Giudici–Spiga [5, Section 2]. In the table below, V is invariant up to S -conjugacy under X . Applying [Lemma 3.3](#), inspection of the indicated factorization table shows that no factor intersection contains V .

We follow the table notation of Xia–Li: P_i is the standard maximal parabolic associated with node i , $P_{i,j} = P_i \cap P_j$ is the intersection of the indicated standard parabolics containing a common Borel subgroup, and N_i denotes the relevant nonsingular i -space stabilizer. The phrase “full $2 + 2$ decomposition stabilizer” includes the element interchanging the two summands.

Socle and conditions	screen subgroup V	table check
$\mathrm{PSL}_2(q)$, excluding the allowed and alternating cases	split-torus normalizer	not in the nonsplit-torus factor of A.1:1 or in P_1 ; A.1:5–8 are checked below
$\mathrm{PSL}_3(q)$, $q \neq 4$, excluding $q = 2$	split-torus normalizer	not in either parabolic or the extension-field factor in A.1:1
$\mathrm{PSL}_3(4)$, every coordinate group X	a pinned maximal subgroup of X	complete Table-of-Marks certificate in Appendix B
$\mathrm{PSL}_n(q)$, even $n \geq 4$	$P_{n/2}$	absent from A.1:1–4 (and A.1:5–10 have other dimensions)
$\mathrm{PSL}_n(q)$, odd $n \geq 5$, no graph	P_2	absent from A.1:1–4,10 except $(n, q) = (5, 2)$
$\mathrm{PSL}_n(q)$, odd $n \geq 5$, graph present	$P_{2, n-2}$	only maximal parabolic overgroups P_2, P_{n-2} ; A.1:1–4,10 are excluded except $(5, 2)$
$\mathrm{PSp}_{2m}(q)$, $m \geq 3$	P_2	absent from A.2:1–16 (A.2:13 has P_2 only when $m = 2$)
$\mathrm{PSp}_4(q)$, q odd	nonsingular $2 + 2$ decomposition stabilizer	absent from the $m = 2$, odd- q rows A.2:1,13
$\mathrm{PSU}_3(q)$	N_1	absent from the $n = 3$ rows A.3:5–7
$\mathrm{PSU}_4(q)$, $q \geq 3$	full nonsingular $2 + 2$ decomposition stabilizer	absent from A.3:1–4,9; in particular A.3:4 covers $\mathrm{PSU}_4(4)$
$\mathrm{PSU}_n(q)$, $n \geq 5$	N_2	absent from A.3:1–4,10–12
$\Omega_{2m+1}(q)$, $m \geq 3$, q odd	P_2	absent from A.4:1–8
$\mathrm{P}\Omega_{2m}^-(q)$, $m \geq 4$	P_2	absent from A.5:1–5
$\mathrm{P}\Omega_{2m}^+(q)$, $m \geq 5$	P_2	absent from A.6:1–13
$\mathrm{P}\Omega_8^+(q)$, triality allowed	central-node P_2	fixed by triality and absent from A.7:1–15

The two corrected orthogonal factorizations have socle intersections N_2^- and a triality image of $\Omega_8^-(q^{1/2})$, for $q = 4, 16$. Neither contains the central-node maximal parabolic P_2 , so they do not alter the last row of the screen.

Here a standard maximal parabolic can only be contained in a proper factor intersection if it is that intersection. A subgroup containing $P_{2,n-2}$ contains a Borel and is therefore parabolic; its only proper overgroups are P_2 and P_{n-2} . These observations justify the containment, rather than mere equality, assertions in the table. The displayed N_i and full $2 + 2$ decomposition stabilizers are standard maximal geometric subgroups for the stated parameters [10, Chapters 3–4]; the exceptional $\text{PSU}_4(2)$ parameter is routed through the low-rank isomorphism below.

For the first row, the split-torus normalizer has order $2(q-1)/(2, q-1)$ and interchanges the two fixed points of its torus, so it is not contained in P_1 . Nor can it lie in the nonsplit-torus normalizer: its order would force $q-1 \mid q+1$, apart from nonsimple parameters. The other A.1 factors occur only for

$$q = 7, 11, 16, 19, 23, 29, 59.$$

The values 7, 11 are allowed. At $q = 19, 29, 59, 23$, divisibility by the torus-normalizer order excludes A_5 or S_4 . At $q = 16$, a subgroup of order 30 lies neither in D_{34} nor in A_5 , the latter because it would have index 2. This closes the rank-one family.

For $S = \text{PSL}_3(q)$, let $d = (3, q-1)$ and let V normalize a maximally split torus. Its class is invariant under diagonal, field, and graph automorphisms. The monomial action shows that V fixes neither a point nor a hyperplane, and

$$|V| = \frac{6(q-1)^2}{d}.$$

The extension-field factor in row 1 of Table A.1 has intersection with S of order $3(q^2 + q + 1)/d$. For $q = 3$, divisibility excludes containment; for $q \geq 5$, the former order is larger. The only remaining table row is the exceptional $q = 4$ row, and all ten almost-simple groups with that socle are handled by the finite certificate. The parameter $q = 2$ gives the allowed group $\text{PSL}_3(2) \cong \text{PSL}_2(7)$.

The even four-dimensional symplectic family is the sole infinite row not closed by the factor screen itself. It is eliminated uniformly by the prime-degree subfield action in [Proposition 4.3](#); no finite calculation is used for that family. The standard low-rank isomorphisms

$$\begin{aligned} \text{PSL}_2(4) &\cong \text{PSL}_2(5) \cong A_5, & \text{PSL}_2(9) &\cong A_6, \\ \text{PSL}_3(2) &\cong \text{PSL}_2(7), & \text{PSL}_4(2) &\cong A_8, \\ \text{PSU}_4(2) &\cong \text{PSp}_4(3). \end{aligned}$$

and the $B_2 = C_2, D_3 = A_3, {}^2D_3 = {}^2A_3$ identifications route every overlap to a row already considered.

A.3. Exceptional groups of Lie type. Hering–Liebeck–Saxl classify all such factorizations [8, Theorems 1–2]. The only socles admitting a core-free factorization are

$$G_2(4), \quad G_2(3^f), \quad F_4(2^f),$$

together with one extra outer factorization for $G_2(4).2$. None of the listed factors is parabolic. For every exceptional socle outside these three families, take V to be a Sylow subgroup in the defining characteristic. Its S -conjugacy class is invariant under every automorphism, and the Hering–Liebeck–Saxl theorem and Lemma 3.3 make the resulting action factor-free.

It remains to screen the three factorable families. If no graph automorphism is present, choose a maximal parabolic. In the graph-fused cases choose the graph-stable parabolic intersection $P_1 \cap P_2$ in $G_2(3^f)$, or $P_1 \cap P_4$ in $F_4(2^f)$; each contains a common Borel. For the graph-fused $G_2(3^f)$ branch, the chosen subgroup has p -part q^6 , whereas the displayed Hering–Liebeck–Saxl factors have p -part at most q^3 . This bound is not being asserted for $G_2(4)$. In the $F_4(2^f)$ branch, the corresponding bounds are at least q^{24} and at most q^{16} . Thus no factor contains the chosen screen subgroup. The small isomorphisms $G_2(2)' \cong \text{PSU}_3(3)$ and ${}^2G_2(3)' \cong \text{PSL}_2(8)$ have already been covered.

A.4. Sporadic groups. Giudici classifies all factorizations with sporadic socle [6, Theorems 1.1–1.3 and Tables 1–4]. Theorem 1.1 and Tables 1–2 treat simple sporadic groups; Theorem 1.2 and Table 3 give the new outer cases; and Theorem 1.3 and Table 4 list the exact factorizations. The factor-free screen uses the complete factorization lists in Tables 1–3, not the exact-only list in Table 4. Comparing those loci with the Atlas maximal-subgroup lists [4, 23] gives the following factor-free maximal subgroups. A row for an outer extension also excludes the factorizations lifted from the simple socle by Giudici’s Lemma 2.1(5).

X	H	X	H
M_{11}	S_5	M_{12}	$2^3:S_4$
$M_{12}.2$	$3^{1+2}:D_8$	$M_{22}.2$	$A_6.2^2$
M_{23}	A_8	J_2	$3.A_6.2_2$
$J_2.2$	$3.A_6.2^2$	HS	$L_3(4).2_1$
$HS.2$	$L_3(4).2^2$	He	$2^2.L_3(4).S_3$
$He.2$	$2^2.L_3(4).D_{12}$	Ru	$2^6:U_3(3):2$
Suz	$J_2.2$	$Suz.2$	$J_2.2 \times 2$
Fi_{22}	$O_7(3)$	$Fi_{22}.2$	$G_2(3).2$
Co_1	$2^{11}:M_{24}$		

The simple group M_{22} , and every sporadic almost-simple group not needed in this display other than M_{24} , has no nontrivial core-free factorization. For such a group, apply Lemma 3.3 to a Sylow subgroup of the socle; its conjugacy class is automorphism-invariant and its maximal overgroup has

nontrivial socle intersection. The rows available in TomLib are separately cross-checked in [Appendix B](#).

The sole factor-cover survivor is M_{24} . Let it act on the 2-subsets of its natural 24-point set. The point stabilizer $M_{22}.2$ is maximal, the degree is 276, and every involution fixes the support of one of its 2-cycles. Thus the action is 2-elusive and [Lemma 3.5](#) applies.

For the eight displayed rows not present in the TomLib cross-check, the maximality assertion is pinned to the “Maximal subgroups” section of the official online ATLAS pages: [He](#), [He.2](#), [Ru](#), [Suz](#), [Suz.2](#), [Fi₂₂](#), [Fi₂₂.2](#), and [Co₁](#).

The four subsections cover every CFSG family, every almost-simple coordinate group, and every low-rank overlap. This completes the detailed audit behind [Proposition 5.1](#).

APPENDIX B. FINITE TABLE-OF-MARKS CERTIFICATES

This appendix records the exact machine interface. The proof uses finite certificates for five small alternating coordinate groups and for the ten almost-simple coordinate groups with socle $\text{PSL}_3(4)$. Nine sporadic rows are also cross-checked.

For a complete table of marks T , let h, c, i represent conjugacy classes of subgroups H, C, I . The command `IntersectionsTom(T,h,c)[i]` is positive exactly when suitable conjugates of H, C have intersection in class i . Such a pair factors X precisely when

$$|H| |C| = |X| |I|.$$

A subgroup C is core-free in an almost-simple X precisely when its class does not contain the socle class. For each pinned named group, the GAP producer locates the unique least nontrivial normal class from the TomLib class lengths and orders; the named almost-simple group identifies that class with its simple socle S . The producer then checks every class not containing S and every intersection class i , verifies $|H \cap S| = |H||S|/|X| > 1$, and fails on missing data. The Python checker validates the pinned inventory, metadata, arithmetic, and file hash, but does not recompute the TomLib intersection predicate.

The proof rows are:

X	$A_{6.2_2}$	$A_{6.2_3}$	$A_{6.2^2}$	A_7	S_7
$ H $	16	20	32	72	144
$ X : H $	45	36	45	35	35.

Since $\text{Out}(\text{PSL}_3(4)) \cong D_{12}$, the following ten TomLib names exhaust, up to isomorphism, the groups between the socle and its full automorphism group:

X	$ H $	$ X : H $	X	$ H $	$ X : H $
$L_3(4)$	960	21	$L_3(4).2_1$	384	105
$L_3(4).2_2$	1920	21	$L_3(4).2_3$	720	56
$L_3(4).3$	216	280	$L_3(4).2^2$	768	105
$L_3(4).3.2_2$	432	280	$L_3(4).3.2_3$	1152	105
$L_3(4).6$	1152	105	$L_3(4).D_{12}$	2304	105

Every row has zero core-free factor classes. The producer and integrity checker are, respectively,

```
gap --quitionbreak -q gap/generate-factor-free-scan.g
python3 tests/check-factor-free-scan.py
```

in the accompanying repository. The environment is GAP 4.15.1 with TomLib 1.2.11. A separate GAP 4.15.1/AtlasRep 2.1.11 regression test reconstructs the corrected $q = 4$ subfield-normalizer instance from [Proposition 4.3](#). All GAP commands use `-quitionbreak`; the Python checkers are also run under `python3 -O`, so their correctness does not depend on assertions. The generated TSV has SHA-256

9b131720d41ef945a0696794c0493ae9c07e166d4d1f37b054a1d55f0c4837ae

and records all twenty-four rows: fifteen proof-critical closeouts and nine sporadic cross-checks. This calculation is an exact finite verification relative to the complete named tables, not an extrapolation from a library scan.

APPENDIX C. DEPENDENCY, SCOPE, AND CORRECTION LEDGER

CFSG. *Exact locus:* the classification theorem. *Consumed statement:* every non-abelian finite simple socle lies in the alternating, Lie-type, or sporadic families used in Appendix A.

Levchuk–Likharev [13]. *Exact loci:* Theorem 1 and Lemmas 4, 6. *Consumed statement:* the three listed simple groups have CMP, which proves occurrence.

Scott [22]. *Exact locus:* the lemma on page 328. *Consumed statement:* a subdirect subgroup of a direct product of nonabelian simple groups is a product of disjoint full diagonal strips.

Liebeck–Praeger–Saxl [17][†]. *Attributed locus:* Corollary 3(iv). *Consumed assumption:* a regular subgroup in the constructed primitive product action forces a core-free transitive coordinate subgroup.

Liebeck–Praeger–Saxl [16][†]. *Attributed locus:* Theorem D and Remark 2. *Consumed assumption:* the large alternating-group factorization screen.

Xia–Li [15] and Gill–Giudici–Spiga [5]. *Exact loci:* Theorem 2.15 and Tables A.1–A.7; Section 2. *Consumed statements:* the readable classical-table restatement and the two corrected orthogonal rows.

Hering–Liebeck–Saxl [8]. *Exact locus:* Theorems 1–2. *Consumed statement:* exceptional Lie-type factorization exhaustiveness.

Giudici [6]. *Exact loci:* Theorem 1.1 and Tables 1–2; Theorem 1.2 and Table 3. *Consumed statements:* the complete simple-sporadic and new outer factorization

lists used for factor-freeness. Theorem 1.3 and Table 4 list only exact factorizations and are not used for that exclusion.

Kleidman–Liebeck [10]. *Source scope:* Chapters 3–4; the group-specific table routing is recorded in Appendix A. *Consumed statement:* maximality of the geometric P_i , N_i , and decomposition-stabilizer screens used in the classical table.

Atlas and online ATLAS [4, 23]. *Exact loci:* the maximal-subgroup lists and the five group-page links after the sporadic table. *Consumed statement:* maximality of the eight published-only sporadic screen subgroups.

Bray–Holt–Roney-Dougal [2] and Burness [3]. *Exact loci:* Table 8.14; Proposition 4.2 and Table 3. *Consumed statement:* maximality of the prime-degree $\mathrm{Sp}_4(q_0)$ subfield subgroup.

Harper [7] and Liebeck–Praeger–Saxl [18]. *Exact loci:* Harper’s setup preceding Lemma 2.1, Lemma 2.1, and Theorem 4(a); LPS Lemmas 2.1–2.2 and Corollary 2.3. *Consumed statements:* the outer automorphisms of $\mathrm{PSP}_4(2^f)$ and its three involution classes.

GAP/TomLib. *Exact environment:* GAP 4.15.1, TomLib 1.2.11; commands in Appendix B. *Consumed output:* fifteen proof-critical factor-free closeouts; the nine sporadic rows are non-proof cross-checks.

Python integrity checker. *Exact program:* `check-factor-free-scan.py`. *Consumed output:* pinned inventory, producer metadata, hashes, and arithmetic. It does not repeat TomLib’s negative intersection search.

GAP/AtlasRep. *Exact environment and program:* GAP 4.15.1, AtlasRep 2.1.11; `test-sp4-subfield.g`. *Consumed output:* none in the theorem; this is a non-proof $q = 4$ regression for the uniform symplectic construction.

The universal conclusion is conditional on CFSG and the exhaustive published classification inputs in this ledger. No property of a section is inferred from CMP; only quotient closure is used. No formal proof assistant is part of the evidence chain.

The daggered LPS loci are load-bearing source inputs. Their exact article texts could not be reopened from an accessible copy during the source traces through 31 August 2026; the bibliographic locators and the available audited-copy hash in the repository’s reference ledger identify the source records. Accordingly, the two consequences formulated in this ledger are explicit external assumptions at the verification boundary. If either original statement has narrower hypotheses, the corresponding product-action or large-alternating branch remains unchecked; no such mismatch is presently known.

Versions 1.0.0 and 1.0.1 used an orthogonal-normalizer assertion that fails for $X = \mathrm{Aut}(\mathrm{PSp}_4(4))$: the old subgroup has index 272, and adjoining the socle gives a subgroup of index 2. The current proof does not use that assertion. It uses the prime-degree subfield subgroup in Proposition 4.3; at the same parameter the corrected subgroup has order 2880, socle intersection 720, and index 1360.

The July 2026 Kourovka Notebook prints Problem 18.68 without a solution annotation. This dated bibliographic observation is not used in the proof and does not rule out unindexed or contemporaneous work.

GENERATIVE-AI DISCLOSURE

Generative-AI tools were used for literature search, proof exploration, code generation, drafting, and critical checking; the author checked and takes responsibility for the manuscript's arguments, citations, and computations.

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